

LiDAR-Based Localisation for Autonomous Vehicles in GPS-Denied Environments Using Dynamic Derivative Analysis

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Abstract—Accurate localisation is a fundamental requirement for the safe and reliable operation of autonomous vehicles (AVs), yet Global Positioning System (GPS) signals remain unreliable or entirely unavailable in urban canyons, tunnels, and densely forested areas. This paper presents a novel LiDAR-based localisation framework that operates independently of GPS by exploiting the rich spatial structure of three-dimensional point cloud data. The proposed method integrates three complementary components: dynamic Savitzky-Golay parameter adjustment for adaptive noise suppression, first- and second-order derivative analysis of polar-transformed point clouds for environmental feature extraction, and a weighted angular-distribution sensor position estimator. Extensive experiments are conducted on the KITTI-CARLA benchmark dataset spanning seven simulated environments that cover urban, rural, and highway scenarios. The proposed approach is compared against four established baseline methods: K-Means clustering, Convex Hull estimation, Minimum Enclosing Circle, and Iterative Closest Point (ICP). Results demonstrate consistent improvements of 57%–96% in localisation accuracy across all environments, with average positional errors as low as 2.46 m, confirming the robustness and generalisability of the framework.

Index Terms—Autonomous vehicles, derivative analysis, GPS-denied localisation, iterative closest point, LiDAR, point cloud processing, Savitzky-Golay filter, sensor localisation.

I. INTRODUCTION

AUTONOMOUS vehicles (AVs) represent one of the most transformative developments in modern transportation, promising to reduce road fatalities, alleviate traffic congestion, and improve mobility for those unable to drive [1]. From simple driver-assistance features such as adaptive cruise control and lane-keeping to the conditional automation defined at SAE Level 3 [2], the field has advanced rapidly over the past decade. Central to this progress is the ability of an AV to determine its precise position and orientation within its environment—a capability known as *localisation*. Without accurate and reliable localisation, even the most sophisticated perception and planning algorithms cannot guarantee safe navigation.

Contemporary localisation systems rely heavily on the Global Positioning System (GPS), which provides metre-level accuracy under open-sky conditions [3]. However, GPS performance degrades significantly in environments where satellite

signals are obstructed or reflected, including urban canyons formed by tall buildings, road tunnels, underground car parks, and densely forested rural roads [17]. In these GPS-denied or GPS-degraded scenarios, AVs must depend on onboard sensors to maintain continuous positional awareness.

Light Detection and Ranging (LiDAR) sensors have emerged as a leading technology for GPS-independent localisation owing to their ability to generate dense, high-resolution three-dimensional (3D) point clouds at high frame rates [4]. Unlike cameras, LiDAR is not affected by ambient lighting conditions, and unlike Radar, it provides centimetre-level spatial resolution, making it well suited for detailed environmental mapping and feature extraction. Despite these advantages, extracting reliable positional estimates from raw LiDAR point clouds in diverse, unstructured environments remains a challenging open problem.

A. Limitations of Existing Approaches

Established LiDAR localisation methods can be broadly categorised into *geometric registration* approaches and *map-based matching* approaches. Geometric registration methods such as Iterative Closest Point (ICP) [5], Generalised-ICP (G-ICP) [6], and Normal Distributions Transform (NDT) [7] align successive point clouds to estimate relative motion. While effective in feature-rich environments, these methods are sensitive to initial alignment quality, prone to local minima, and computationally demanding for real-time deployment. Map-based methods [10], [11] achieve high accuracy but require pre-built high-definition (HD) maps that are costly to construct and maintain, and which quickly become outdated in dynamic environments.

Simpler geometric estimators—including K-Means centroid clustering, Convex Hull approximation, and Minimum Enclosing Circle fitting—offer lower computational cost but oversimplify the spatial distribution of point cloud data. These methods frequently fail to capture fine-grained environmental structure, leading to large positional errors in scenarios involving tight turns, path overlaps, or varying point cloud density.

B. Contributions

This paper addresses these limitations by proposing a novel LiDAR-only localisation framework that does not require HD maps, prior scene knowledge, or GPS assistance. The framework processes raw LiDAR point cloud data through a three-stage pipeline that adaptively suppresses noise, extracts

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meaningful environmental boundaries, and triangulates sensor position using angular feature density. The specific contributions of this work are as follows.

- 1) **Dynamic Savitzky-Golay Adaptive Filter:** A parameter-free smoothing stage that automatically selects the optimal window length and polynomial order for the Savitzky-Golay filter [23] based on the noise characteristics of each individual point cloud frame, preserving critical structural features while suppressing measurement noise.
- 2) **Polar-Domain Derivative Analysis:** A novel application of first- and second-order derivatives to polar-transformed LiDAR data that identifies significant environmental boundaries—such as edges, corners, and surface discontinuities—which serve as reliable localisation anchors across diverse scene types.
- 3) **Weighted Angular Position Estimator:** A sensor position estimation function that refines an initial geometric-centroid estimate by weighting contributions from angularly dense regions of the point cloud and further correcting the estimate using derivative-identified feature locations.
- 4) **Comprehensive Experimental Evaluation:** Systematic evaluation on the KITTI-CARLA benchmark dataset across seven distinct simulated environments, demonstrating improvements of 57%–96% in localisation accuracy over four baseline methods: K-Means, Convex Hull, Minimum Enclosing Circle, and ICP.

C. Paper Organisation

The remainder of this paper is structured as follows. Section II reviews related work in LiDAR-based localisation. Section III describes the KITTI-CARLA dataset and the baseline localisation methods used for comparison. Section IV presents the proposed framework in detail. Section V reports experimental results and analysis. Section VI concludes the paper and outlines directions for future research.

II. RELATED WORK

Localisation for autonomous vehicles has been studied extensively across a range of sensor modalities. This section reviews the most relevant prior work, organised by sensing approach, and identifies the gaps that motivate the proposed method.

A. LiDAR-Based Localisation

LiDAR sensors have been a cornerstone of autonomous vehicle research since the DARPA Urban Challenge [1], owing to their ability to generate dense, geometrically accurate 3D point clouds independent of lighting conditions. The most widely adopted LiDAR localisation algorithms are built on point-cloud registration.

Iterative Closest Point (ICP) [5] estimates the rigid transformation between successive scans by iteratively minimising the sum of squared distances between corresponding points. Although accurate in feature-rich environments, ICP is susceptible to local minima, sensitive to the quality of the initial

alignment estimate, and computationally expensive for dense point clouds. *Generalised-ICP (G-ICP)* [6] extends ICP by modelling each point as a Gaussian distribution, improving robustness to noise but increasing computational cost. *Normal Distributions Transform (NDT)* [7] represents the environment as a set of Gaussian distributions on a voxel grid and aligns scans by maximising the likelihood of observed points. Takeuchi and Tsubouchi [8] extended NDT to 3D, while Akai *et al.* [9] incorporated experimentally determined scan uncertainty to further improve robustness.

Map-matching approaches represent a complementary strategy. Levinson and Thrun [10] demonstrated precise urban localisation by matching intensity measurements against a pre-built probabilistic map. Wolcott and Eustice [11] employed multiresolution Gaussian mixture models, achieving sub-metre accuracy across diverse weather conditions. Burgard *et al.* [12] proposed a map-based framework tailored to urban environments by exploiting permanent structural features. Although effective, all map-based methods require costly offline map construction and are vulnerable to map staleness in dynamic environments.

Yoneda *et al.* [13] demonstrated that matching salient scan features against a high-precision 3D map yields accurate localisation but noted that performance degrades in feature-sparse scenarios such as open highways. Im *et al.* [14], [15] proposed vertical corner and line map representations to reduce map size while maintaining localisation accuracy in urban areas. Li *et al.* [16] introduced a hybrid filtering framework that combines particle filtering with LiDAR scan matching, improving robustness in industrial environments with repetitive structure.

In contrast to these methods, the proposed framework requires no pre-built map, no initial pose estimate, and no iterative scan registration, instead extracting position directly from the spatial statistics of a single point cloud frame.

B. GPS and Inertial Navigation

GPS remains the dominant positioning technology for outdoor vehicles due to its global coverage and ease of integration [3]. However, multipath effects and signal blockage in urban environments, tunnels, and under dense tree cover severely degrade GPS accuracy [17]. Inertial Navigation Systems (INS), based on accelerometers and gyroscopes, can bridge short GPS outages through dead-reckoning but suffer from unbounded drift accumulation over time [18]. Tightly-coupled GPS/INS fusion partially mitigates this by continuously correcting inertial drift, yet the fundamental dependence on satellite availability persists.

Aldibaja *et al.* [19] demonstrated intensity-based LiDAR localisation as a GPS substitute on snow-wet road surfaces, highlighting LiDAR's particular advantage in adverse weather and motivating the GPS-free philosophy adopted in this paper.

C. Radar and Vision-Based Approaches

Radar-based localisation has attracted growing interest due to its robustness in adverse weather. Ward and Folkesson [20] demonstrated localisation using low-cost automotive radar,

TABLE I
COMPARISON OF REPRESENTATIVE LOCALISATION METHODS

Method	Map-free	GPS-free	Reg.-free	Single-frame
ICP [5]	✓	✓		
NDT [7]	✓	✓		
Map-matching [10]		✓		
GPS/INS [18]	✓		✓	✓
Radar [20]	✓	✓		
Proposed	✓	✓	✓	✓

while Gwon *et al.* [21] generated lane-level road maps suitable for radar-based guidance. However, radar provides significantly lower spatial resolution than LiDAR, limiting its ability to resolve fine environmental structure.

Vision-based localisation leverages the rich semantic content of camera images for place recognition, lane detection, and landmark matching [22]. While camera systems offer high resolution at low cost, their performance is highly sensitive to illumination changes, lens soiling, and adverse weather—conditions under which LiDAR continues to operate reliably.

D. Point Cloud Filtering and Feature Extraction

Preprocessing raw LiDAR data is a critical step in any localisation pipeline. The Savitzky-Golay filter [23], originally developed for analytical chemistry, has been adopted in robotics and sensor signal processing for its ability to smooth discrete noisy sequences while preserving relative maxima, minima, and overall signal shape—properties essential for maintaining the geometric fidelity of LiDAR returns. Crucially, the selection of window length and polynomial order governs the trade-off between noise suppression and feature preservation; fixed parameters inevitably mismatch the noise characteristics of individual frames in dynamic driving scenarios. The proposed method addresses this by dynamically optimising these parameters on a per-frame basis using mean squared error minimisation.

E. Summary and Positioning

Table I summarises the key characteristics of representative localisation methods. The proposed approach occupies a distinct niche: it is map-free, GPS-independent, registration-free, and operates on a single point cloud frame without any prior scene information.

III. DATASET AND BASELINE METHODS

This section describes the benchmark dataset used for evaluation and provides a concise formulation of the four baseline localisation methods against which the proposed framework is compared.

A. The KITTI-CARLA Dataset

Experiments are conducted on the *KITTI-CARLA* dataset [24], [25], a synthetic benchmark that combines the data

Fig. 1. Ground-truth sensor trajectories for Towns 01–07 of the KITTI-CARLA dataset. Each town exhibits distinct path geometry, ranging from closed urban loops (Towns 01–03) to open highway segments (Towns 06–07).

format and annotation conventions of the KITTI vision benchmark [24] with scenes generated by the CARLA open-source urban driving simulator [25]. CARLA provides physically realistic sensor models, including a 64-beam LiDAR sensor that generates dense point clouds at 10 Hz, and produces ground-truth pose information with millimetre-level precision—enabling rigorous quantitative evaluation of localisation accuracy.

The dataset spans **seven distinct environments** (Town 01 through Town 07), each representing a different type of driving scenario:

- *Towns 01–03*: Urban environments with multi-lane roads, intersections, roundabouts, and dense roadside infrastructure.
- *Towns 04–05*: Mixed environments combining urban districts with open suburban roads and varying building density.
- *Towns 06–07*: Rural and highway-style environments with fewer structural landmarks, wider lane separations, and reduced point cloud density.

Each town provides a sequence of LiDAR point cloud frames stored in PLY format, paired with ground-truth sensor poses expressed as (x, y) coordinates in a local Cartesian reference frame. The diversity of environments—in terms of structural density, road curvature, and point cloud characteristics—ensures that evaluation results are not biased towards any single scene type. A visualisation of the ground-truth trajectories for all seven towns is shown in Fig. 1.

B. Data Preprocessing

Prior to applying any localisation method, each raw PLY point cloud undergoes two preprocessing steps common to all methods evaluated in this work.

Outlier removal: Statistical outlier filtering is applied using an Isolation Forest model to identify and discard points whose Euclidean distance from the local neighbourhood mean

exceeds three standard deviations. This step removes spurious returns caused by dust particles, sensor noise, and distant reflections that carry no useful structural information.

Coordinate projection: The 3D point cloud (x, y, z) is projected onto the horizontal plane by discarding the z -component, yielding a 2D point set $\mathcal{P} = \{(x_i, y_i)\}_{i=1}^N$ suitable for planar localisation. This is consistent with standard practice in urban driving, where lateral positioning is the primary concern.

C. Baseline Localisation Methods

Four geometric baseline methods are evaluated. All operate on the same preprocessed 2D point set $\mathcal{P}$ described above and produce a scalar position estimate $\hat{\mathbf{p}} = (\hat{x}, \hat{y})$.

1) *K-Means Centroid (K-Means Path):* K-Means clustering [4] partitions $\mathcal{P}$ into k clusters by iteratively minimising the within-cluster sum of squared Euclidean distances. The sensor position is estimated as the centroid of the largest cluster, on the assumption that the densest spatial cluster corresponds to the region nearest the sensor:

$$\hat{\mathbf{p}}_{\text{KM}} = \frac{1}{|\mathcal{C}^*|} \sum_{(x_i, y_i) \in \mathcal{C}^*} (x_i, y_i), \quad (1)$$

where $\mathcal{C}^*$ denotes the largest cluster. In this study $k=3$ is used, chosen empirically to balance cluster granularity and stability. K-Means offers low computational cost but is sensitive to initialisation and to the uneven spatial distribution of LiDAR returns, particularly in open environments.

2) *Convex Hull Centroid (Convex Hull Path):* The convex hull of $\mathcal{P}$ is the smallest convex polygon enclosing all points. The sensor position is estimated as the centroid of this polygon:

$$\hat{\mathbf{p}}_{\text{CH}} = \frac{1}{M} \sum_{j=1}^M \mathbf{v}_j, \quad (2)$$

where $\mathbf{v}_1, \dots, \mathbf{v}_M$ are the M vertices of the convex hull. While robust to noise by construction, the convex hull oversimplifies complex, non-convex environments and is insensitive to the internal distribution of points, leading to systematic bias in scenarios with asymmetric point density.

3) *Minimum Enclosing Circle (Minimum Circle Path):* The minimum enclosing circle (MEC) is the smallest circle that contains all points in $\mathcal{P}$. The sensor position is taken as its centre $\hat{\mathbf{p}}_{\text{MEC}} = (c_x, c_y)$, computed via Welzl's algorithm in expected $O(N)$ time. Like the convex hull, the MEC is dominated by the most distant outlier points and fails to exploit the density distribution of returns, producing estimates that mirror the convex hull errors in curvature-rich scenes.

4) *Iterative Closest Point (ICP Path):* ICP [5] estimates the rigid transformation $(\mathbf{R}, \mathbf{t})$ that aligns the current point cloud $\mathcal{P}_t$ to a reference cloud $\mathcal{P}_{t-1}$ by minimising:

$$E(\mathbf{R}, \mathbf{t}) = \sum_i \|\mathbf{R} \mathbf{p}_i^{(t)} + \mathbf{t} - \mathbf{q}_i^{(t-1)}\|^2, \quad (3)$$

where $\mathbf{q}_i^{(t-1)}$ is the closest point in $\mathcal{P}_{t-1}$ to the transformed point $\mathbf{R} \mathbf{p}_i^{(t)} + \mathbf{t}$. The accumulated translation vector is used

Fig. 2. Overview of the proposed three-stage localisation pipeline. A raw LiDAR point cloud is converted to polar coordinates, adaptively smoothed, differentiated to identify environmental features, and finally used to produce a refined sensor position estimate.

to update the global position estimate. ICP is the most sophisticated of the four baselines but is the most sensitive to initialisation quality: a poor initial alignment causes convergence to a local minimum, resulting in large positional errors that propagate across frames.

D. Evaluation Metric

All methods are evaluated using the *mean Euclidean error* (MEE) between the estimated sensor position $\hat{\mathbf{p}}$ and the ground-truth position $\mathbf{p}^*$ across all T frames in a town sequence:

$$\text{MEE} = \frac{1}{T} \sum_{t=1}^T \|\hat{\mathbf{p}}_t - \mathbf{p}_t^*\|_2. \quad (4)$$

The percentage improvement of the proposed method over a baseline b is then defined as:

$$\Delta_b = \frac{\text{MEE}_b - \text{MEE}_{\text{proposed}}}{\text{MEE}_b} \times 100\%. \quad (5)$$

IV. PROPOSED METHOD

The proposed localisation framework processes a single preprocessed 2D point cloud $\mathcal{P} = \{(x_i, y_i)\}_{i=1}^N$ through three sequential stages: (i) adaptive noise suppression via a dynamically parameterised Savitzky-Golay filter, (ii) environmental feature extraction via polar-domain derivative analysis, and (iii) weighted angular position estimation. Fig. 2 illustrates the complete pipeline.

A. Stage 1: Polar Coordinate Transformation

The Cartesian point set $\mathcal{P}$ is first converted to a polar representation centred on the geometric mean $\bar{\mathbf{p}} = (\bar{x}, \bar{y})$:

$$r_i = \sqrt{(x_i - \bar{x})^2 + (y_i - \bar{y})^2}, \quad \theta_i = \text{atan2}(y_i - \bar{y}, x_i - \bar{x}), \quad (6)$$

Algorithm 1 Dynamic Savitzky-Golay Parameter Selection

Input: signal $\{s_i\}_{i=1}^N$, max window $w_{\max}$
Output: optimal window w^* , polynomial order d^*
best_MSE $\leftarrow \infty$
for each odd $w \in \{11, 13, \dots, w_{\max}\}$ **do**
 for $d \in \{2, 3, 4\}$ **do**
 Apply Savitzky-Golay filter: $\tilde{s}_i \leftarrow \text{SG}(s, w, d)$
 MSE $\leftarrow \frac{1}{N} \sum_i (\tilde{s}_i - s_i)^2$
 if MSE < best_MSE **then**
 best_MSE $\leftarrow$ MSE; $w^* \leftarrow w$; $d^* \leftarrow d$
 end if
 end for
end for
return w^*, d^*

yielding the polar set $\mathcal{Q} = \{(r_i, \theta_i)\}_{i=1}^N$. The data is then sorted by θ_i in ascending order to produce a continuously ordered angular sequence, which is a prerequisite for meaningful derivative computation in the next stage.

B. Stage 2: Dynamic Savitzky-Golay Adaptive Smoothing

Raw LiDAR returns contain measurement noise that, if uncorrected, causes spurious extrema in derivative signals and corrupts feature detection. The Savitzky-Golay filter [23] smooths a discrete sequence by fitting overlapping polynomial windows via least squares, preserving signal shape characteristics such as local maxima and minima. For a window of length w and polynomial order d , the smoothed value at position i is:

$$\tilde{s}_i = \sum_{j=-m}^m c_j s_{i+j}, \quad m = \frac{w-1}{2}, \quad (7)$$

where c_j are the Savitzky-Golay convolution coefficients determined by the least-squares polynomial fit and $s_i \in \{r_i, \theta_i\}$.

Dynamic parameter selection. The key limitation of the standard Savitzky-Golay filter is that the window length w and polynomial order d must be chosen in advance. Fixed parameters that suit one frame may over-smooth structural features or under-suppress noise in another. To address this, we introduce an automatic parameter optimisation procedure that selects w^* and d^* independently for each frame by minimising the mean squared error (MSE) between the smoothed signal and the original:

$$(w^*, d^*) = \arg \min_{w \in \mathcal{W}, d \in \{2, 3, 4\}} \frac{1}{N} \sum_{i=1}^N (\tilde{s}_i^{(w,d)} - s_i)^2, \quad (8)$$

where $\mathcal{W} = \{11, 13, 15, \dots, w_{\max}\}$ is the set of odd window lengths up to a maximum $w_{\max}$ (set to $\min(51, N)$ in practice to avoid over-smoothing small point clouds). This optimisation is performed separately for both the radial sequence $\{r_i\}$ and the angular sequence $\{\theta_i\}$, yielding two independently tuned filters. Algorithm 1 summarises the procedure.

C. Stage 3: Polar-Domain Derivative Analysis

With the smoothed radial and angular sequences $\{\tilde{r}_i\}$ and $\{\tilde{\theta}_i\}$ in hand, we compute first- and second-order finite differences to locate regions of significant environmental change.

1) First-Order Derivatives: The first-order derivatives quantify the rate of change of range and angle with respect to the sorted sample index i :

$$r'_i = \frac{\tilde{r}_{i+1} - \tilde{r}_{i-1}}{2}, \quad \theta'_i = \frac{\tilde{\theta}_{i+1} - \tilde{\theta}_{i-1}}{2}. \quad (9)$$

Large magnitudes in $|r'_i|$ indicate abrupt changes in measured range, corresponding to surface discontinuities such as building edges, kerbs, and vegetation boundaries.

2) Second-Order Derivatives: The second-order derivatives detect inflection points and regions of rapidly accelerating change, which correspond to corners and concave features:

$$r''_i = \tilde{r}_{i+1} - 2\tilde{r}_i + \tilde{r}_{i-1}, \quad \theta''_i = \tilde{\theta}_{i+1} - 2\tilde{\theta}_i + \tilde{\theta}_{i-1}. \quad (10)$$

3) Angular Wrap-Around Correction: Because $\theta \in (-\pi, \pi]$ is a cyclic variable, naive finite differences produce spurious large values at the $\pm\pi$ boundary. A correction is applied wherever $|\theta'_i| > \pi$:

$$\theta'_i \leftarrow \theta'_i - 2\pi \cdot \text{sign}(\theta'_i), \quad (11)$$

ensuring derivative continuity across the angular wrap-around.

D. Stage 4: Weighted Angular Position Estimation

The final stage fuses the geometric centre, the angular density distribution, and the derivative feature map into a refined sensor position estimate through three successive refinement steps.

Step 1 — Geometric centroid initialisation. An initial position estimate is computed as the mean of all point coordinates:

$$\hat{\mathbf{p}}^{(0)} = \frac{1}{N} \sum_{i=1}^N (x_i, y_i). \quad (12)$$

Step 2 — Angular density refinement. The angular distribution $\{\theta_i\}$ is discretised into a histogram $H(\phi)$ with bin width $\Delta\phi = 5^\circ$. Peaks in $H(\phi)$ identify angular directions where point cloud returns are most densely concentrated, corresponding to nearby planar surfaces such as walls or building facades. The position estimate is shifted towards the weighted mean of these dominant angular directions:

$$\hat{\mathbf{p}}^{(1)} = \hat{\mathbf{p}}^{(0)} + \alpha \sum_{k \in \mathcal{K}} H(\phi_k) \begin{pmatrix} \cos \phi_k \\ \sin \phi_k \end{pmatrix}, \quad (13)$$

where $\mathcal{K}$ is the set of peak bin indices and α is a normalisation constant scaled by the total point count.

Step 3 — Derivative-based feature correction. Points with high combined derivative magnitude:

$$D_i = |r'_i| + |r''_i| + |\theta'_i| + |\theta''_i| \quad (14)$$

are identified as environmental feature anchors (edges, corners, discontinuities). The top $\kappa = 10\%$ of points ranked by D_i form the feature set $\mathcal{F}$. The final estimate is computed as the weighted centroid of $\mathcal{F}$, with weights proportional to D_i :

$$\hat{\mathbf{p}}^{(2)} = \frac{\sum_{i \in \mathcal{F}} D_i (x_i, y_i)}{\sum_{i \in \mathcal{F}} D_i}. \quad (15)$$

The output $\hat{\mathbf{p}}^{(2)}$ is the final sensor position estimate returned by the framework.

TABLE II
MEAN EUCLIDEAN ERROR (MEE, METRES) AND IMPROVEMENT OF THE PROPOSED METHOD OVER BASELINES ACROSS ALL SEVEN KITTI-CARLA TOWNS

Town	Environment	Baseline MEE (m) / Improvement (%)				Proposed MEE (m)
		K-Means	Conv. Hull	Min. Circle	ICP	
Town 01	Urban	11.60 / 68.9	27.70 / 87.0	27.70 / 87.0	73.34 / 95.1	3.61
Town 02	Urban	9.41 / 70.2	23.30 / 88.0	23.30 / 88.0	65.08 / 95.7	2.80
Town 03	Urban	7.51 / 58.5	23.54 / 86.8	23.54 / 86.8	73.99 / 95.8	3.12
Town 04	Mixed	11.14 / 72.4	32.18 / 90.4	32.18 / 90.4	73.87 / 95.8	3.08
Town 05	Mixed	7.98 / 57.0	31.07 / 89.0	31.07 / 89.0	58.27 / 94.1	3.43
Town 06	Rural/Highway	9.55 / 69.3	26.56 / 89.0	26.56 / 89.0	75.32 / 96.1	2.93
Town 07	Rural/Highway	8.82 / 72.0	23.16 / 89.4	23.16 / 89.4	67.32 / 96.3	2.46
Mean		9.43 / 66.9	26.78 / 88.5	26.78 / 88.5	69.60 / 95.6	3.06

E. Computational Complexity

The dominant cost is the dynamic parameter search in Stage 2. For a point cloud of N points, $|\mathcal{W}|$ candidate window lengths, and three polynomial orders, the complexity is $O(|\mathcal{W}| \cdot N)$ per signal. Sorting in Stage 1 costs $O(N \log N)$, and all derivative and estimation steps are $O(N)$. The total per-frame complexity is therefore $O(|\mathcal{W}| \cdot N)$, which is linear in N for a fixed search grid—substantially cheaper than the iterative $O(N^2)$ worst-case cost of ICP.

V. EXPERIMENTAL RESULTS

This section presents quantitative and qualitative evaluation of the proposed framework against the four baseline methods across all seven KITTI-CARLA towns. All experiments are implemented in Python using NumPy, Pandas, Scikit-learn, and Plyfile, and executed on a standard desktop CPU without GPU acceleration.

A. Quantitative Results

Table II reports the mean Euclidean error (MEE, in metres) for each method in each town, together with the percentage improvement Δ_b (eq. (5)) of the proposed method over each baseline. The proposed method consistently achieves the lowest MEE in all seven environments.

B. Analysis by Baseline Method

1) *K-Means Path*: K-Means produced the lowest baseline errors among the four methods, with MEE ranging from 7.51 m (Town 03) to 11.60 m (Town 01), reflecting its ability to broadly capture dominant spatial clusters in the point cloud. However, K-Means is inherently sensitive to cluster initialisation and to uneven point density caused by partial occlusions. The proposed method improved upon K-Means by 57.0%–72.4% across towns, with the smallest gains in feature-sparse rural environments (Town 05: 57.0%) and the largest in mixed and structured urban scenes (Town 04: 72.4%). These gains arise because the derivative analysis extracts sub-cluster boundary structure that K-Means centroids cannot resolve.

2) *Convex Hull and Minimum Circle Paths*: The Convex Hull and Minimum Enclosing Circle methods produced identical MEE values in every town, a consequence of both methods being dominated by the same set of outermost points in the 2D projection. MEE ranged from 23.16 m to 32.18 m, consistently three to four times worse than K-Means. Both methods fail to exploit the internal spatial distribution of the point cloud, reducing the localisation problem to a geometric bounding operation that is highly sensitive to outlier points near the scan periphery. The proposed method improved upon these methods by 86.8%–90.4%, the most consistent improvement band observed across any baseline.

3) *ICP Path*: ICP produced the largest errors of all baselines, with MEE ranging from 58.27 m (Town 05) to 75.32 m (Town 06), representing roughly 6–8 $\times$ the error of the proposed method. The catastrophic failure of ICP in this setting is attributable to the absence of a reliable initial alignment: without a prior pose estimate, ICP frequently converges to incorrect local minima and the accumulated registration errors compound rapidly across frames. The proposed method improved over ICP by 94.1%–96.3%, the highest improvement margins observed, confirming that single-frame derivative-based estimation is far more stable than multi-frame registration in GPS-denied cold-start scenarios.

C. Cross-Environment Consistency

A notable characteristic of the proposed method is its consistent performance across environments of very different character. The MEE varies only between 2.46 m (Town 07, rural/highway) and 3.61 m (Town 01, dense urban), a range of just 1.15 m despite the substantial differences in structural density and road geometry between these environments. This contrasts with the baselines, which exhibit much wider variance: K-Means MEE spans 4.09 m and ICP MEE spans 17.05 m across the same towns.

The relatively lower error in rural and highway towns (06–07) compared to dense urban towns (01–03) reflects the effect of sparser, more regular point cloud structure: fewer overlapping returns from nearby buildings means that derivative peaks more reliably identify genuine environmental boundaries rather than cluttered near-field returns.

Fig. 3. Estimated sensor trajectories vs. ground truth for Town 01 (urban, top) and Town 07 (rural/highway, bottom). The proposed method (solid blue) closely follows the ground truth (dashed black), while ICP (red), Convex Hull (green), and K-Means (orange) diverge significantly.

Fig. 3 illustrates the estimated trajectories for all four baselines and the proposed method overlaid on the ground-truth path for representative Towns 01 and 07. The proposed method tracks the ground truth closely throughout, while ICP diverges almost immediately and the geometric methods (Convex Hull, Minimum Circle) produce trajectories offset by tens of metres.

D. Error Distribution Analysis

Fig. 4 shows per-frame error plots for all methods on Town 01. The proposed method maintains a low and stable error throughout the sequence, with occasional short-duration spikes at sharp turns where derivative features are momentarily ambiguous. K-Means exhibits moderate but volatile errors with frequent excursions above 10 m. The Convex Hull and Minimum Circle methods cluster between 30–40 m with little frame-to-frame variation, indicating systematic rather than random bias. ICP errors cluster around 75–80 m with high variance, consistent with accumulated registration drift.

E. Improvement Summary

Fig. 5 summarises the percentage improvement of the proposed method over each baseline across all towns. The most consistent improvements are over ICP (94–96%) and the geometric bounds methods (87–90%). Improvement over K-Means is somewhat lower (57–72%) but still substantial, confirming that the proposed derivative-based feature extraction provides significant accuracy gains even over the strongest baseline.

VI. CONCLUSION AND FUTURE WORK

A. Conclusion

This paper presented a novel GPS-independent LiDAR localisation framework for autonomous vehicles, designed to operate reliably in environments where satellite positioning is

Fig. 4. Per-frame localisation error (metres) for all methods on Town 01. The proposed method (blue) maintains low stable errors throughout the sequence. K-Means (orange) is volatile; Convex Hull and Minimum Circle (green/purple) show systematic bias; ICP (red) suffers severe accumulated drift.

Fig. 5. Percentage improvement of the proposed method over each baseline method across all seven KITTI-CARLA towns.

unreliable or entirely unavailable. The framework processes a single 3D LiDAR point cloud frame through a four-stage pipeline: polar coordinate transformation, dynamic Savitzky-Golay adaptive smoothing, first- and second-order polar-domain derivative analysis, and weighted angular position estimation. Unlike map-based or scan-registration methods, the proposed approach requires no pre-built HD map, no initial pose estimate, and no inter-frame correspondence, making it suitable for cold-start deployment in previously unseen environments.

Comprehensive evaluation on the KITTI-CARLA benchmark across seven environments—spanning dense urban, mixed suburban, and rural/highway scenarios—demonstrated consistent and substantial improvements over four established baselines. The proposed method achieved a mean Euclidean error of 3.06 m averaged across all towns, representing im-

improvements of 57%–72% over K-Means clustering, 87%–90% over Convex Hull and Minimum Enclosing Circle estimation, and 94%–96% over Iterative Closest Point registration. Critically, the method maintained low and stable per-frame errors throughout each town sequence, in contrast to ICP which suffered severe accumulated drift and the geometric bounding methods which exhibited persistent systematic bias.

The dynamic Savitzky-Golay parameter selection proved particularly effective at adapting the noise suppression level to the varying density and noise characteristics of individual frames, while the derivative-based feature identification reliably located structural boundaries—edges, corners, and surface discontinuities—that serve as robust localisation anchors across diverse scene types.

These results establish the proposed framework as a practical, computationally efficient foundation for GPS-denied localisation in autonomous vehicles, with relevance extending to mobile robotics, unmanned aerial vehicles, and infrastructure-free mapping systems.

B. Limitations

The current work has several limitations that bound the scope of the results. First, all experiments are conducted on simulated data from the KITTI-CARLA dataset. Although CARLA provides high-fidelity sensor models, the gap between simulated and real sensor noise, weather effects, and dynamic object interference remains to be characterised. Second, the framework currently operates in 2D by projecting the point cloud onto the horizontal plane; tall or multi-level structures whose vertical extent is informative for localisation are therefore not exploited. Third, the method produces absolute position estimates in a local coordinate frame but does not accumulate these estimates into a globally consistent trajectory, meaning long-term drift correction is not addressed.

C. Future Work

Several directions offer promising opportunities to extend this work.

Real-world validation. The most immediate priority is evaluation on real LiDAR datasets such as the original KITTI odometry benchmark [24] and the nuScenes dataset, which include real sensor noise, dynamic agents, and varying weather conditions. Such evaluation will reveal whether the performance gains observed in simulation transfer to practical deployment.

3D extension. Extending the derivative analysis to the full 3D point cloud by operating in spherical coordinates (r, θ, ϕ) would allow the framework to exploit vertical structural information—particularly useful in multi-storey urban environments and underground car parks where horizontal projections are ambiguous.

Temporal fusion and trajectory estimation. Integrating successive single-frame position estimates via a Kalman filter or particle filter would suppress frame-to-frame noise and enable globally consistent trajectory estimation. Coupling this with wheel odometry or IMU data would further improve

robustness during featureless stretches such as long straight highways.

Sensor fusion. Fusing LiDAR-derived position estimates with camera-based semantic landmarks, radar returns, and GPS when available would create a resilient multi-modal localisation system that degrades gracefully as individual sensors fail or become unreliable.

Deep learning integration. Learned feature detectors trained on large-scale LiDAR datasets could replace the hand-crafted derivative-based feature extractor, potentially improving robustness to sensor noise and point cloud sparsity. A hybrid approach—using learned features for initial coarse localisation and derivative analysis for refinement—is a particularly promising direction.

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